Anudeep Nayak

github.com/Anudeep-Nayak | 9945830497 | anudeep.nayak@protonmail.com
Linkedin.com/Anudeep-Nayak/ | anudeepnayak.dev

EDUCATION

MS in Data Science — August 2023 - May 2025
University of Colorado Boulder — **GPA 3.93**

B. Tech in Computer Science and Engineering — August 2018 - June 2022
Manipal Institute of Technology

SKILLS

Languages & Database: Python, Java, Go, C++, JavaScript, TypeScript, SQL, MySQL, ArangoDB, AQL, R, HTML, CSS, Redis
Tools & Technology: Git, GitHub, Agile, CI/CD, Jenkins, REST APIs, AWS, Azure Cloud, IBM Cloud, GCP, Docker, Kafka

PROFESSIONAL EXPERIENCE

Software Engineer — October 2025 - Present
Actualize — Boulder, Colorado
- Architected Aether Chat, a **context-aware AI** conversational agent that processes **historical user behavior** and **geospatial data** to dynamically generate personalized, location-based event recommendations
- Integrated **Redis** in-memory caching solution to optimize database queries and API responses, achieving up to **50%** improvement in system performance metrics
- Implemented **RabbitMQ** message queue system with graceful shutdown and startup mechanisms, reducing average processing wait times by over **500ms** while ensuring **zero message** loss during deployments

Graduate Scholarship Assistant — September 2024 - April 2025
University of Colorado - Boulder — Boulder, Colorado
- Streamlined scholarship tracking system by automating data processing workflows using **Python** and **Pandas**, eliminating manual entry tasks and saving over **288 hours** annually
- Created interactive **HTML/JavaScript** templates that dynamically render personalized scholarship requirements during counseling sessions

Software Engineer — July 2022 - August 2023
GE Healthcare — Bangalore, India
- Developed and deployed over **20** software features using **Java, Node** and **SQL** on **Azure Cloud**, emphasizing quality and scalability in a collaborative **Agile** team environment
- Collaborated directly with stakeholders to integrate authentication **APIs**, streamlining supply chain data entry and reducing processing time by **80%**
- Resolved **40+** bugs annually, leveraging **CI/CD** and source control (**Git**) tools to enhance system stability and minimize downtime by **20%**

Software Engineering Intern — January 2022 - June 2022
GE Renewable Energy — Bangalore, India
- Modernized legacy user interfaces by executing a full migration from **Angular 6** to **Angular 11** (**TypeScript**, **HTML**, **CSS**); leveraged **Jenkins** CI/CD on **AWS** to streamline deployments and successfully delivered **12+** user stories within six months
- Enhanced code quality through **unit testing** and cleanup, increasing test coverage from **45%** to **80%** and reducing duplication by **50%**
- Restructured a ratings page to capture **NPS** as a **KPI** index. The redesign (**UX**) improved customer experience and increased number of ratings by **10%**

PROJECT EXPERIENCE

AniTA network (AI TA) — University of Colorado Boulder
- Architected an AI teaching assistant (**Django, Docker, GCP**) utilizing native **AQL** graph analytics in **ArangoDB**; leveraged **Jaccard Similarity** for grading consistency and the **Louvain Method** to cluster student learning gaps.
- Achieved **90%** automated grading accuracy by integrating the **Claude API** to evaluate student submissions, while reducing **AI hallucinations** by **87%** through a **PageRank-driven RAG** pipeline that retrieves and supplies historical feedback.

GE Renewable Energy Hackathon — GE Renewable Energy
- Engineered a machine learning categorization system using **scikit-learn** and the **KNN algorithm** to parse and classify resumes, achieving an **83%** accuracy rate.
- Awarded **1st place** overall by optimizing model performance through the implementation of novel data filtering, preprocessing, and text mining techniques.

Dota 2 LLM chatbot — University of Colorado Boulder
- Built a domain-specific conversational agent utilizing the **IBM Watson API** to process and resolve complex in-game Dota 2 queries with context-aware responses.
- Achieved **90%** response accuracy by architecting a **Retrieval-Augmented Generation** (**RAG**) pipeline that ingested, vectorized, and retrieved unstructured data from comprehensive gaming encyclopedias.